

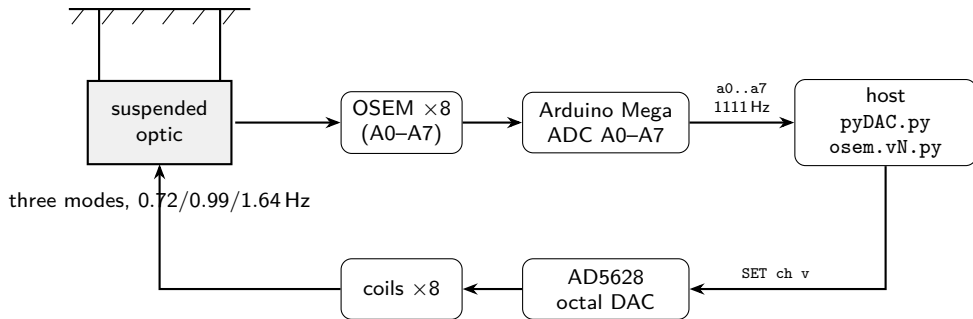
OSEMa bin-laden

Fourteen rungs of a suspended-optic damping loop: v0 to v13

7 August 2026

Eight OSEM shadow sensors, eight coils, one suspended mirror,
and fourteen attempts at the supervisor around them.

What is on the bench

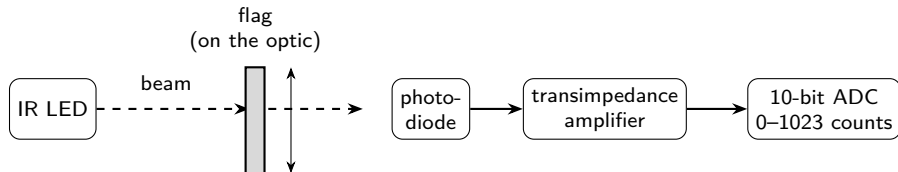


Sensor $i \rightarrow$ pin $A_i \rightarrow$ DAC_CHANNELS = [1,3,5,7,0,2,4,6].

All eight OSEMs and all eight coils are connected.

Four of them, a0-a3, carry usable in-band signal and their four coils push back.

The OSEM: a shadow sensor



- ▶ A flag on the optic partly occludes the beam. **The photodiode current is linear only while the flag is partly in the beam.** Fully blocked or fully clear, the sensor reports nothing about motion.
- ▶ **The ADC reads 0 to 1023 counts.** Its two ends are the *rails*.
- ▶ **Two voltage scales, never to be conflated.** Sensing: $A_{VCC} = 5.02\text{ V}$ over 1023 counts, so **4.907 mV/count**. Actuation: DAC full scale 2.5 V, of which the loop uses V_{MIN} , $V_{MAX} = 0.0, 0.5\text{ V}$ around $BIAS = 0.25$.
- ▶ **At a rail the sensor is outside its linear region**, so anything derived from it is wrong until the flag comes back.

The plant: three resonances, and a bound on the damping

A suspended optic is a damped harmonic oscillator. For one sensed coordinate $x(t)$, mass m , force $F(t)$, and mode m :

$$\ddot{x} + \frac{\omega_m}{Q_m} \dot{x} + \omega_m^2 x = \frac{F(t)}{m}$$

Amplitude decays as $e^{-\gamma_m t}$, so

$$\gamma_m = \frac{\omega_m}{2Q_m}, \quad \tau_m \equiv 1/\gamma_m.$$

γ is per mode. There are three of them.

Resonances, 320 s ringdown, coils at bias:

$$f_A, f_B, f_C = 0.7155, 0.9949, 1.6396 \text{ Hz}$$

$$\text{each } \pm 0.0001 \text{ Hz}$$

The per-mode γ_m fits are noise-dominated.

Across channels, in s^{-1} :

	ch	median	range
A	4	0.0483	0.0107–0.1388
B	5	0.0082	0.0005–0.0579
C	4	0.0094	0.0061–0.0530

Mode A spans a **factor of 13** across four channels watching one rigid body. **Do not quote a per-mode γ as measured.**

What is established is *aggregate*, from a 12 s autocorrelation exclusion test over all channels and the in-band signal as a whole. The fitted $\gamma_{\text{agg}} = +0.00004 \pm 0.00719 \text{ s}^{-1}$ is consistent with zero:

$$Q > 433 (1\sigma) \Rightarrow \tau > 138 \text{ s}$$

One bound over all three modes, not three measurements.

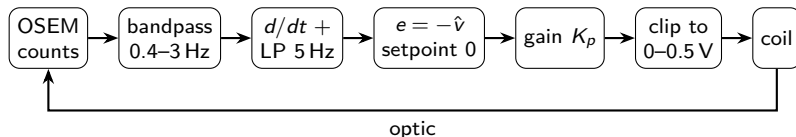
Closed loop the same kick settles in **2.9–4.7 s**.

Cold damping: the process variable is *velocity*

$$\text{bp}(t) = \text{LP}_{3\text{ Hz}} \left[\text{HP}_{0.4\text{ Hz}}[A(t)] \right]$$

$$\hat{v}(t) = \text{LP}_{5\text{ Hz}} \left[\frac{\text{bp}(t) - \text{bp}(t - \Delta t)}{\Delta t} \right]$$

$$u(t) = -K_p(t) \cdot \hat{v}(t)$$



The setpoint is a constant zero, so $e = -\hat{v}$. The loop does not ask where the optic *is*, only that it stop *moving*. All three measured modes sit inside 0.4–3 Hz.

Three gains: P is a damper, I is a spring, D is a mass

With $e = -\hat{v}$ and $\hat{v} \approx \dot{x}$, the velocity of the sensed coordinate x :

$$\underbrace{K_p e = -K_p \dot{x}}_{\text{force} \propto \text{velocity}}$$

$$\underbrace{K_i \int e dt = -K_i x}_{\text{force} \propto \text{displacement}}$$

$$\underbrace{K_d \frac{de}{dt} = -K_d \ddot{x}}_{\text{force} \propto \text{acceleration}}$$

term	acts on	physically	effect on the pole
K_p	velocity	viscous damper	the only term that removes energy
K_i	displacement	added spring	moves the resonance; does not damp
K_d	acceleration	added (negative) mass	noisiest; differentiates twice

Why add I and D at all? K_p sets the damping, but it leaves a steady offset it cannot remove and it reacts only to present error. K_i takes out the offset; K_d anticipates. Neither dissipates, so **amplitude ratio is the wrong metric to expect an improvement in.**

The velocity gains, and what set them

The loop feeds back velocity. For OSEM i , $\hat{v}_i$ is its bandpassed signal differentiated (**V/s**) and u_i is the voltage added to that channel's coil bias (**V**): $u_i = -K_p^{(i)} \hat{v}_i$, so K_p is in **V per V/s**.

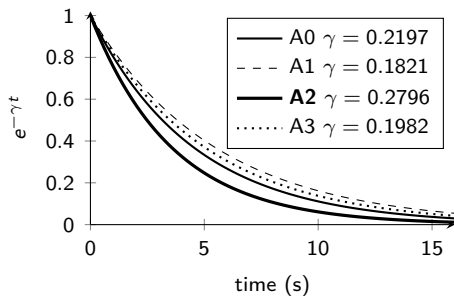
	ch0	ch1	ch2	ch3	ch4-7
K_p	-.035	-.035	+.035	-.035	0
K_i	-.0060	-.0060	+.0060	-.0060	0
K_d	-.00045	-.00045	+.00045	-.00045	0

K_p is **negative** because the coil force must *oppose* velocity.

ch2 is positive and that is correct, not a typo. Its OSEM and coil are mounted the other way round, so its loop sign is inverted: $K_p > 0$ there opposes velocity as $K_p < 0$ does elsewhere.

- ▶ $|K_p| = 0.030$ was set against an oscilloscope on ch0: **6× amplitude reduction, no rail clipping**. The only value in the project fixed that way.
- ▶ **Bound: $|K_p| = 0.040$ is the measured onset of instability and riling.** Do not exceed 0.035 **unattended**, which is where the shipping value sits.
- ▶ K_i and K_d take each channel's sign, because all three drive one coil through one loop. Their *magnitudes* are set by the actuator budget: peak coil demand **0.2001 V** inside a ± 0.25 V window.
- ▶ **ch4–ch7 run at zero:** their coil sign is not determined, and a wrong-signed channel *pumps*.
- ▶ Anti-windup is back-calculation, TRACK_TC_S = 0.5 s; I_CLAMP_V = 0.15 V is a backstop that has never engaged (measured peak $|I|$ **0.084 V**).

The evidence that ch2's sign is inverted



Envelopes $e^{-\gamma t}$ at the *measured* γ ($R^2 = 0.65\text{--}0.92$), all four channels damping together.

Fit $e^{-\gamma t}$ to each channel's decay with all four damping. A2 has the **largest** γ of the four, while running at $K_p = +0.010$ where ch0/ch1/ch3 ran at -0.030 .

A positive K_p on a normally-wired channel *pumps*. The only reading that fits: **ch2's coil or OSEM is mounted the other way round**.

Confirmed three times. On the bench ch2 ran positive alongside three negative channels and came out best, ratio 0.01–0.03 against ch0's 0.08. And in a fixed-frequency refit ch2's residue is **the only positive one of the four, in both modes**: +13.573 against $-7.851 / -8.867 / -5.282$ on mode A.

This is not a bug and must never be "corrected". The simulator encodes it as $\text{COIL_GAIN}[2] = -4.15$.

The gain schedule, and the baseline everything scales off

$$\text{RMS}_{\text{base}} = \sqrt{\frac{1}{N} \sum \text{bp}(t_i)^2} \quad \text{at gain} = 0$$

$$r(t) = \text{RMS}_{\text{win}}[\text{bp}(t)] / \text{RMS}_{\text{base}}$$

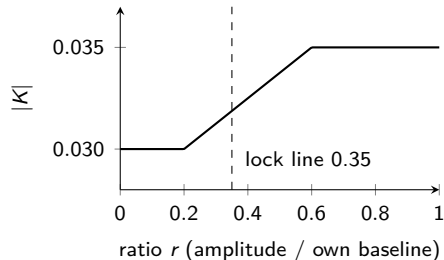
$$K(t) = K_{\text{steady}} + s(r) (K_{\text{cap}} - K_{\text{steady}})$$

$$s(r) = \text{clip}\left(\frac{r-0.2}{0.4}, 0, 1\right)$$

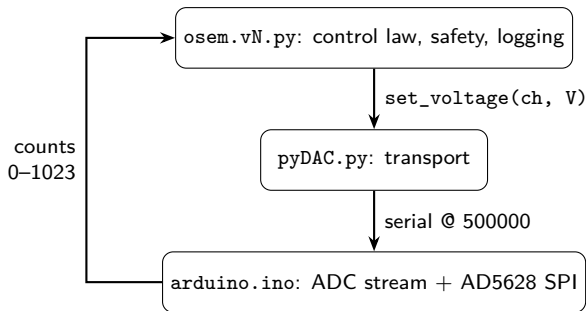
Slew-limited at $\text{GAIN_SLEW_PER_S} = 0.02$, so no gain step can itself kick a transient. The same ramp is the startup soft-start: $0 \rightarrow \text{capture gain}$ in 1.75 s.

One number, three jobs. The runaway breaker trips above $1.8\times$ baseline. The scheduler blends between $0.2\times$ and $0.6\times$. LOCKED means below $0.35\times$ held for 5 s.

Everything downstream of calibration is a ratio against this one number, measured once, in one window.



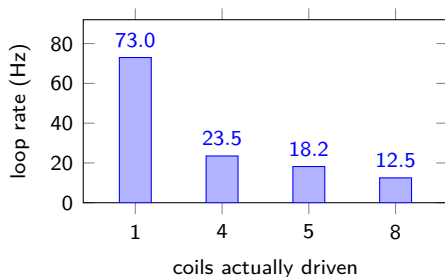
Three layers, one wire



direction	message	meaning
MCU → host	READY	sent once after setup()
host → MCU	SET <ch> <volts>	sets DAC 0-7; replies OK ch=.. v=..
host → MCU	STREAM / STOP	replies STREAMING / STOPPED
MCU → host	a0,...,a7	free-running ADC counts, one line per loop

Opening the port toggles DTR, resets the board and zeroes all eight DAC channels: **connecting is itself a safing action**. **OK ch=.. v=.. shares the wire with the data rows**, and nothing in the framing distinguishes an ack from a sample row.

The wire is the bottleneck, and the loop rate falls with coil count



Median DAMPING-row spacing over 20 bench logs: 13.7
/ 42.6 / 55.0 / 79.9 ms.

Same run, both states: **357.1 Hz** while CALIBRATING
with no coil writes against **18.2 Hz** while DAMPING on 5
coils. **A factor of 19.6**, with no between-run variation
to argue about.

set_voltage blocks reading up to 50 stream lines
waiting for its OK. **Every coil write costs a serial
round trip**, so the loop rate is set by how many
coils are driven, not by the control law.

Exactly one threshold was counted in *samples*
rather than seconds, so its meaning moved with
the coil count:

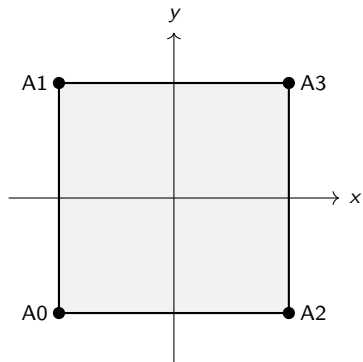
MAX_CONSECUTIVE SATURATED = 30

coils	1	4	5	8
means	0.41 s	1.28 s	1.65 s	2.40 s

Less sensitive exactly as the rig got harder to
control.

And it costs phase. Margin at 3.75 Hz: **73.5°** on 4
coils, **45.5°** on 8, **105.0°** once the ack wait goes.
Every gain in the rig was fixed on a 4-coil loop.

Four sensors, one rigid body



$$z_i = \text{LONG} + \text{PITCH} \cdot y_i + \text{YAW} \cdot x_i$$

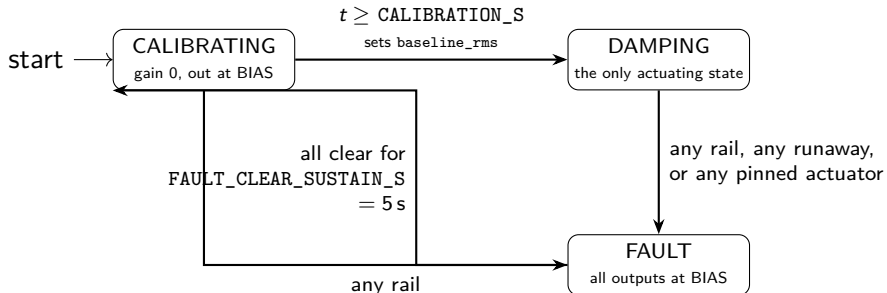
mode	z_0	z_1	z_2	z_3
LONG	+	+	+	+
PITCH	-	+	-	+
YAW	-	-	+	+
BUTTERFLY	+	-	-	+

Four sensors, three rigid-body freedoms. BUTTERFLY is identically zero for a rigid body, so any reading in it is one sensor disagreeing with the other three.

Two measured consequences the project rests on. One OSEM/coil pair damps the *whole* mass, ≈ 70 s. Four together settle it in ≈ 17 s.

That $4\times$ is why v1 exists, and it is why a quorum of *one* healthy channel is enough.

The supervisor: three states



No transition out of DAMPING except into FAULT, and no terminal state. Ctrl+C leaves through a finally block that returns every DAC channel to bias.

What does *not* run in FAULT: the gain schedule, the runaway check and the lock update. Any flag those set on the way in is therefore never recomputed while the fault is held.

v0: the baseline

ENABLE_CHANNEL = [True, False, False, False]

P-only: $u = -K \cdot v$, KI_GAIN = KD_GAIN = 0.

What v0 got right, and every version since inherits:

1. Runs continuously, not as fixed-duration sweep steps.
2. Measures its *own* baseline every restart.
3. A rail interlock on *raw counts*, separate from the RMS breaker. A blind OSEM flatlines, which an RMS-only check reads as perfect stability.
4. Auto-recovery out of FAULT.
5. A lock detector that prints a real number.
6. Streaming CSV, flushed periodically, survives an unclean kill.
7. Gain scheduling with a slew-limited ramp.

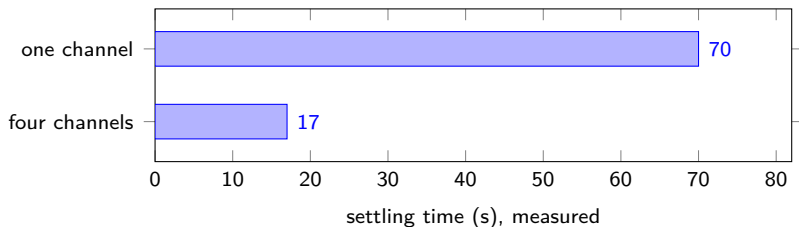
v0, through harness.py

ch0 amplitude ratio	0.188
time to lock	12.1 s
suite (when present)	18/18
undriven ch1/2/3	0.333 / 0.316 / 0.253

Three defects shipped with it, present in v0, v1 and v2: saturation-latch, rail-threshold, runaway-baseline.

Plus a fourth nobody found until v9, and a fifth until v12.

v1: one line changed



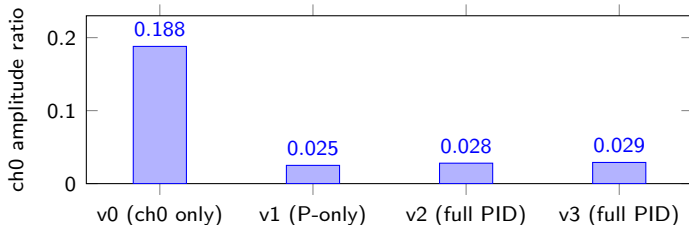
`ENABLE_CHANNEL = [True]*4`. Nothing else moved: same gains, filters, safety code, same three defects.

v0's docstring claim that only ch0 was validated was **already two days out of date**. §4 of the report documents all four damping simultaneously.

Why 4×, when it is still four SISO loops? The four loops act *in parallel* over the coupled modes, so the overall rate is closer to the *sum* of their contributions than to the slowest single coupling path.

Measured: ch0 ratio 0.188 → **0.025**, lock 12.1 → **10.7 s**.

v2: I and D switched on, and the prediction that held



I and D brought nothing in damping, as predicted, because neither dissipates. On hardware v2 was *worse*: its integrator was non-zero in **63%** of samples and pushed the pinned-actuator streak from **28 to 31** against a threshold of **30**.

Peak $|I|$ was 0.084 V against a 0.150 V clamp, so **I_CLAMP_V never engaged**. Bounded is not the same as unwinding, which is why v3 replaces the clamp with back-calculation.

On the bench, 2026-08-03: the only hardware evidence for v1–v3

	v1	v2	v3
Law	P-only	full PID	full PID
Lock time	13.8 s	13.9 s	14.5 s
Baseline (ch0)	0.6316 V	0.3158 V	0.1804 V
Peak bp	3.141 V	2.837 V	0.410 V
ADC clipped	2.19%	10.40%	0.00%
Samples with non-zero /	0 of 6939	4709 of 7513	all
Longest pinned run	28	31	0
Trip type	runaway	saturation	none
Outcome	recovered in 5.0 s	latched ~50 s	no fault

Do not compare ratio across these runs. It is normalised to each run's own baseline and those differ by $3.5\times$. Normalising the absolute locked RMS gives 0.139/0.128/0.115: indistinguishable. **These runs rank failure behaviour, not damping quality.**

Defect 1: saturation-latch

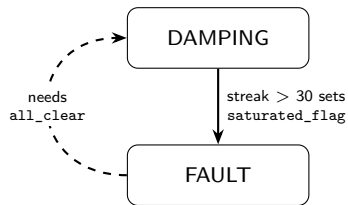
The data. v2 on the bench, ch0 output against the $V_{MAX} = 0.5\text{ V}$ rail:

u (V)	sat_streak
0.4987	0
0.5000	1
⋮	⋮
0.5000	30
0.5000	31

v1 reached **28** and never tripped. v2's integrator added the three samples that crossed **30**.

The smoking gun. Across **16,966** FAULT samples v2 logged **one** distinct ratio value: **3.6803**.

What goes wrong. saturated_flag is recomputed only inside check_runaway(), which does not run in FAULT.



The flag can never clear: permanent fault.

A runaway trip recovered fine. **Only saturation deadlocked.**

The fix, and the number that moved. v3 maintains the flag in actuate(), so it refreshes every sample in every state. In FAULT the output is at bias, so it clears itself.

v2 latched ~50s. **v3, same bench: 0 pinned samples, 0.00% ADC clipped, no fault.**

Defect 2: rail-threshold, an interlock at the noise floor

The data. v0's test: raw counts below
RAIL_LOW_COUNTS = 3 of 1023, i.e. **14.7 mV**, for
250 consecutive samples.

A blind channel with ~ 1 count of dark noise:

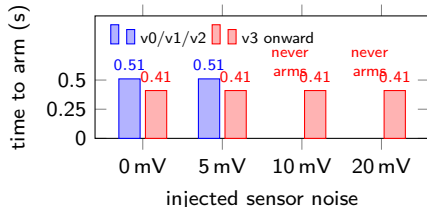
counts	consecutive count
0 1 2 0 1 5	
0 2 1 3 0 10	
4	$\rightarrow 0$
1 0 2	3

One sample above 3 counts erases everything.
The channel is blind the whole time.

On hardware: v1 logged **246** genuinely clipped
ADC samples (2.19% of the run) and **the rail flag**
never set once.

What goes wrong. Above ~ 1 count RMS of
dark noise the interlock **stops arming**. A blind
channel's bandpassed output flatlines, so its
ratio goes to zero: it reads as the **steadiest axis**
on the rack while the loop declares itself locked
on a dead sensor.

The fix, and the number that moved. v3:
threshold 3 $\rightarrow$ **12 counts**, and the test becomes
80% of a 0.5 s window instead of an unbroken
run. Noise now *dilutes* the evidence instead of
erasing it.



Defect 3: runaway-baseline, and what an inflated floor costs

The data. Three runs, same rig, same afternoon, ch0 calibrated baseline:

	v1	v2	v3
baseline (V)	0.6316	0.3158	0.1804
runaway line, $1.8\times$	1.137	0.568	0.325
lock line, $0.35\times$	0.221	0.111	0.063

A $3.5\times$ spread in the floor is a $3.5\times$ spread in every threshold derived from it. v1 could call itself LOCKED at 0.221 V of residual motion where v3 needed 0.063 V.

Absolute locked RMS, normalised:

0.139/0.128/0.115. The three runs were *equally* well damped. Only the yardstick moved.

What goes wrong. The floor is one RMS over one 8 s window at zero gain. A bump inside that window inflates it for the whole run, which desensitises the breaker and makes the lock claim easy to satisfy.

The fix, and the number that moved. v3: calibrate 20 s, take the **median of 5 sub-windows**, so one bad sub-window is outvoted. Worst shock skew 39.3% $\rightarrow$ **19.7%** against a 35% assertion line (simulator, one 6 V/s impulse swept across the window).

Improved, not solved. Gain is zero during calibration, so a shock rings down over a timescale comparable to the window and there is no clean sub-window left to prefer. The code labels it a partial fix.

The completion is v12's baseline-sanity, and it took nine more versions.

v3: the control law becomes a real PID

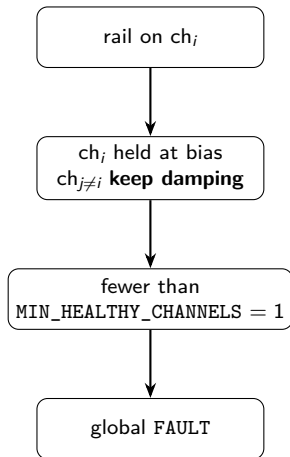
change	why
derivative on <i>measurement</i> , not on error	identical while the setpoint is a constant zero, but it cannot produce a derivative kick if that ever changes
back-calculation anti-windup , <code>TRACK_TC_S = 0.5 s</code>	the old scheme froze the accumulator at a hard limit; this bleeds it back at a defined time constant, so recovery from saturation does not depend on which side of a conditional the error fell on
one <code>reset()</code> , on every entry to a non-actuating state	exactly one definition of bumpless re-entry

Two kinds of change kept separate: **the control law** above, **the three defect fixes** on the previous slides. A bench comparison can then attribute an effect to one or the other.

`Channel.p_term / i_term / d_term / ki / kd` survive as *properties*, because the logger, the status line and the simulator's live gain editing use those names.

v3 is the minimal PID baseline to A/B against, and one of four controllers with `BENCH_STATUS = validated`.

v4: auto-disable, demote the axis, do not freeze the rig



The problem. v3 freezes all four channels on any one rail. One OSEM losing its flag stops *all* damping, **at the moment of largest excursion**, which is what made the flag leave range.

A quorum of one is not a guess: the v0 campaign measured one OSEM/coil pair damping the whole rigid body.

condition	v4	why
rail	demote	local <i>sensing</i> failure: that measurement is garbage
runaway	global	not localisable: four loops, one body
saturation	global	measurement fine, only the output clipped

Saturation is deliberately *not* demoted: a saturated channel is still pushing the right way at full authority.

First fired on hardware under v10: ch2 railed, 3 of 4 kept damping, re-armed 2 s later.

v5: four channels become four lanes

	total	code	logic	main()
v3	908	515	400	64
v4	1172	577	460	64
v5	533	356	270	48

Channel went from **231 lines to 13**. It stopped being where state lives and became a *view* onto the controller's arrays, so `sim/server.py` and `harness.py` kept working unchanged.

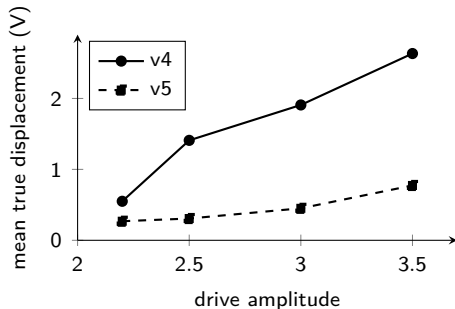
This is what made eight channels a config change rather than a rewrite.

Evidence it is the same controller. Every printed observable matches v4 exactly, on two different plants:

observable	v4, and v5 identically
ratios	0.029 / 0.098 / 0.045 / 0.132
lock	10.6 s
peak $P/I/D$	0.0881 / 0.0644 / 0.0412 V
re-arm	2.10 s

One regression, in the direction you would not guess: $57.5 \mu\text{s}/\text{step}$ against v4's 42.4. numpy's per-call overhead on a length-4 array beats four scalar operations. Both are $\sim 50\times$ inside the 2.88 ms budget, so it is irrelevant on hardware.

v5: fast-refault, stop re-measuring during the event



Simulator *ground truth* over 300s, not the controller's own estimate.

The problem. v0–v4 answer every fault the same way: wait for all-clear, then recalibrate. That is CALIBRATION_S of open loop at zero gain. Correct for a transient, wrong for a disturbance that has not gone away: the optic rings straight back up to the amplitude that faulted it.

200 s @ drive 2.5	% DAMPING	faults	locks
v0 / v1 / v2	0.6%	1	never
v3 / v4	4.1%	7	never
v5	87.5%	1	10.2 s

The fix is four lines. If the previous engagement lasted under FAST_REFAULT_S = 15s, the cause is still present, so re-engage on the baseline already in hand. Bounded by MAX_BASELINE_REUSE = 4.

Found by stressing the simulator across clipping regimes, not by reading the code.

v5.5: eight OSEMs, a config change rather than a rewrite

v5 with $N = 8$ and the full map `DAC_CHANNELS = [1,3,5,7,0,2,4,6]`. **Not a new rung**: same control law, constants, `FIXES` and state machine.

All eight sit on one rigid body, so every semantic held unchanged: quorum of one, one global `FAULT`, one lock claim.

Channels 4–7 shipped disabled. Their sign was unknown, and a wrong-signed channel *pumps*. `ch2`'s sign was *discovered* from bench data, not predicted.

It never announced LOCKED, and it faulted the whole rig **ten times in 145 s**.

The surviving four were not comfortably damping either. Bench, eight channels:

t	ch0	ch1	ch2	ch3
135.3 s	0.71	0.68	0.81	0.78
145.4 s	0.13	0.12	0.05	0.08

At $t = 135.3$ s the gain was still ramping from zero for the third time (-0.0148 of -0.030). They reached LOCK only on the last status line of the run, and **ch5 never did**, so the rig itself never claimed LOCKED.

ch5 never showed LOCK at all, and the rig-level claim needs *every* live channel. That quorum rule is still unfixed, and it is open defect 1.

Defect 4: a near-zero baseline makes noise look like a runaway

The data. v5.5 calibrated baselines on the bench:

	a0	a4	a5	a6	a7
baseline (V)	0.3109	0.0014	0.0801	0.0050	0.0020
1.8×	0.560	0.0025	0.144	0.0090	0.0036

a4's own bandpassed output never left ± 0.01 V.

Against a trip line of **0.0025 V** that is **4×** over.

Measured peak ratios: a4 **3.98**, a6 **4.79**, a7 **11.31**.

All on a signal too small to see.

What goes wrong. Two properties combine.

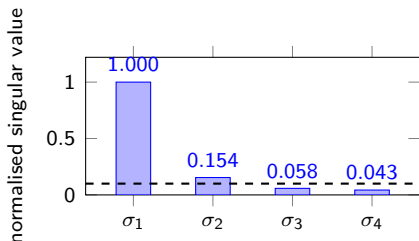
1. An OSEM with no in-band signal **does not rail**. It sits mid-scale with a flat bandpassed output, so auto-disable never fires.
2. Runaway is a **global** fault, because it is not localisable.

So a channel measuring its own noise takes the entire rig down: **ten whole-rig faults in 145 s, nine of them naming a4, a6 or a7.**

The fix, and the number that moved. v10's baseline-floor. At the end of every calibration, demote any channel whose baseline is under **10% of the median** across enabled channels.

2026-08-06: a4/a6/a7 demoted at $t = 20.001$ s, then **40 s of unbroken DAMPING and zero faults** on the same eight OSEMs.

v6: measuring the actuation matrix properly



Dashed: the 10% retention line. **Directions above**
it: 2

KIND = "sysid". **Not a controller**: it damps nothing and closes no loop.

Stepped sine, one coil at a time. While coil j drives, no other coil is. Nothing to disentangle, no matrix to invert, no assumption about quadrature: **magnitude and phase both fall out of the lock-in.**

Only 4 of 12 frequencies were usable: ch2 railed in **17 of 48** steps, and two survivors are ambient-dominated.

v6.1 (multisine) failed outright: ch2 railed in **all 4** passes, and pass 1 ran at **14.8 Hz** effective against ~ 252 Hz, giving a spurious ~ 4700 V/V against ~ 70 .

One cause for both: open loop, nothing damping, ch2 clips. **And the rank check is superseded**: the ringdown finds **three** modes, and the full refit puts the allocator at 3×4 , cond **1.41**.

The DC actuation matrix, and the conclusion it will not support

	coils 0–3				coils 4–7			
a0	−105	−107	−48	+19	−29	−21	−28	−26
a1	−49	−68	−33	+18	+5	+7	+4	+8
a2	+35	+66	+213	−141	−25	−22	−13	−26
a3	+36	+14	+58	−51	−1	−2	+2	−1
a4	+0	+2	+2	+1	+8	+8	+2	+1
a5	+4	+0	+0	−6	−0	−7	−1	+8
a6	+1	+4	+2	+3	+3	+3	+4	+6
a7	−0	+3	+1	+0	+1	−1	+6	+7

$d(\text{counts}_i)/d(\text{bias}_j)$, one coil stepped at a time, bench

2026-08-04.

Block means of response		
	coils 0–3	coils 4–7
sensors 0–3	66	14
sensors 4–7	2	4

Coils 4–7 work: they move a0–a3 by ~ 25 counts/V. **Sensors a4–a7 respond to nothing,** ≤ 8 counts/V.

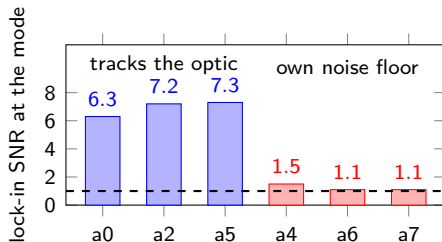
We read that as “a4–a7 are not wired”. It was **wrong**. A DC slope against a fixed threshold cannot separate *no sensor* from *a working sensor with small gain*. **Wrong statistic, not wrong data.**

Their own diagonals are **+8/ +4/ +7** against a block mean of **4**. Those signs are coin flips, and **a wrong sign pumps**, so v11 and v12 ship a4/a6/a7 at **zero gain**.

a3: the common-mode sweep read $+43$ counts/V, coil3 $\rightarrow$ a3 alone -51 .

Common-mode slopes are not per-coil slopes.

All eight *are* wired: in-band against out-of-band



Lock-in on the two long 8-channel logs (118 s, 148 s)
at the mode frequencies. Dashed: SNR = 1.

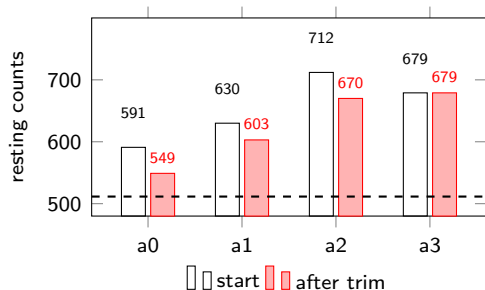
All eight OSEMs are electrically connected, confirmed on a scope. The split is *in band* against *out of band*:

power in 0.4–3 Hz		
a0–a3	77–99%	the three modes
a5	22–71%	in band here, not in the residue fit
a4, a6, a7	0.1–9%	a narrow 6.19 Hz line (12.4 Hz harmonic on a7)

Not an Arduino or sampling problem. a3 is read *immediately before* a4 and carries a large 1.64 Hz signal, yet $r(a4, a3) = -0.029 / +0.004$. a5, read straight *after* a4, is clean.

The cause of the low gain is NOT established. “Flags outside the linear region” is **unsupported**: a4 rests at **567.5** counts against mid-scale **511.5** with a **30-count swing**, nowhere near an end.

v7: bias-trim, move where the optic hangs



Dashed: ADC mid-scale, 511.5 counts. Bench 2026-08-04. a3 was already at its best step.

Coil bias is a **DC force**, so stepping it moves where the optic hangs. One `BIAS_QUANTUM` = 0.25 V at a time, kept only if the **total** offset across all channels improved.

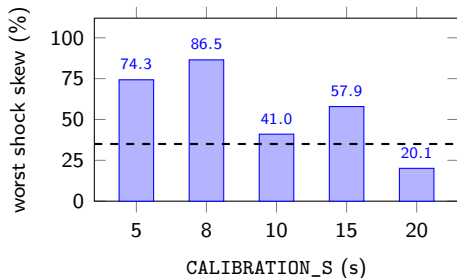
bench	2026-08-04	start	end
v7 (trim)		565	455 counts
v5 (control)		566	546 (drift)

Lock times 11.2 / 11.0 / 12.4 s: **the trim costs nothing.**

Two steps were rejected and frozen by the *total* test: ch1 → 0.50 V (486 → 528) and ch3 → 1.25 V (538 → 561). Each helped its own channel and hurt the rig, which is the inter-coil coupling v6 measured.

It cannot finish the job. Centring wants $-1.93 \dots + 3.77$ V against a 2.5 V DAC. The rest is mechanical or TIA offset.

v8: stop paying 20 s of zero gain for a baseline



Dashed: the 35% assertion line. **Every shorter window fails**, some worse than the unfixed defect (39.3%).

The cost. Bench, four channels: 20 s calibration plus ~ 11 s to lock is **65% of wall clock at zero gain**, plus 25 s of open loop after every fault.

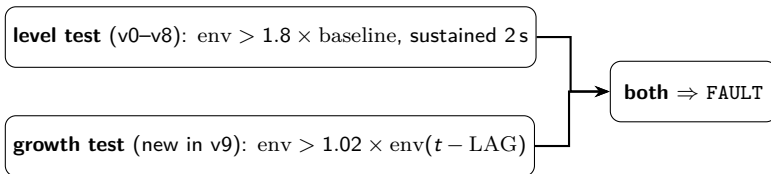
1. fast-calib. CALIBRATION_S becomes a *ceiling*. Sub-windows accumulate; calibration ends as soon as the trailing ones agree the floor has stopped moving. Simulator: 20.00 $\rightarrow$ **6.01** s, lock 30.6 $\rightarrow$ **16.5** s, and shock skew *improves* 19.7 $\rightarrow$ 17.6%.

2. warm-restart. A fault after a *healthy* engagement re-engages on the baseline in hand. Two 15 V/s kicks: v4/v5 never re-lock inside 115 s; v8 re-engages 5 s after all-clear and re-locks 10.7 s later.

On the bench fast-calib never fired.

CALIB_AGREE_TOL = 1.20 was tuned on simulator noise, one RNG seed. It degrades to v5's estimator, which is the designed fallback, but the saving is **unconfirmed on hardware**.

Defect 5: the runaway breaker was asking the wrong question



A **runaway** is the loop pumping energy *in*: the envelope **grows**.

A **ringdown** is a large but **falling** envelope: the loop *succeeding*.

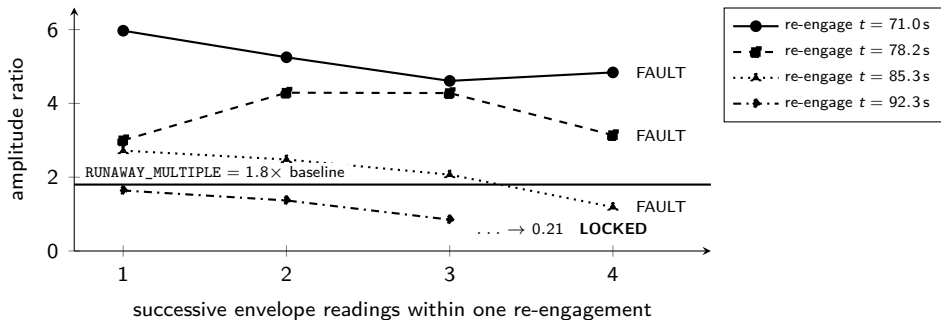
A level test cannot tell them apart. After a kick it trips on a timer, every RUNAWAY_SUSTAIN_S, all the way down the ringdown.

This defect was in every version back to v0. v8 merely re-engaged fast enough to hit it repeatedly, which is how it was found.

RUNAWAY_GROWTH_FRAC = 1.02 gives 2% of headroom so envelope noise alone cannot read as growth.

RUNAWAY_TREND_LAG_S = 2.0 was chosen before the mode frequencies were known. That turns out to be the next defect, and it took three more versions to find.

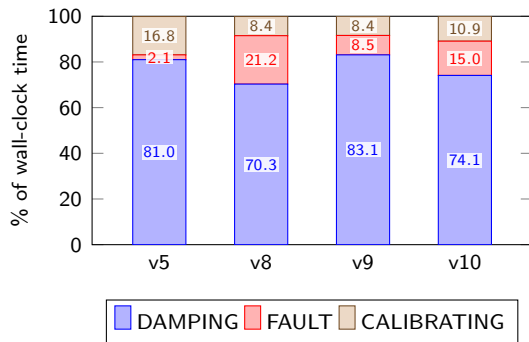
v9: the bench evidence, the envelope *falling* through every fault



Measured on **v8, hardware, 240 s, three kicks: ten faults**, and the envelope was *falling* through them.

Only the re-engagement that happened to start at 1.64, just under the line, survived.

v9: what the growth gate bought



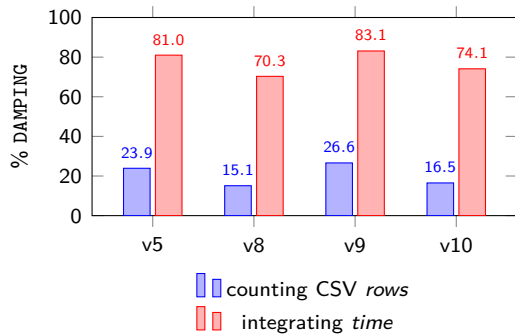
same 3-kick schedule	faults	locks
v5 (control)	1	2
v8	10	4
v9	4	4
v10 (08-06)	8	3

v9 re-locked after every kick, at $t = 31.3/83.8/151.3/205.3$ s. v5's low fault time is not calm: it locked **2 of 4**.

v9's FAULT time is **20.2 s** =
FAULT_CLEAR_SUSTAIN_S 5 s $\times$ 4
recoveries. **Cost, not thrash.**

v9 still holds the best DAMPING fraction on record, 83.1%. Every later comparison has to say what schedule it ran.

Defect 6: two estimators, one inverted conclusion



Identical CSVs and state column. Only the weighting differs.

The data. v10's **74.1%** (time) against v9's **26.6%** (rows) = $2.8\times$.

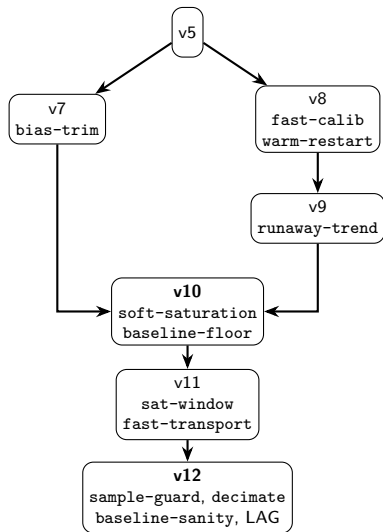
What goes wrong. That reads as “v10 nearly tripled the damping duty”. The two halves came from different estimators. **Like for like, either way, v9 wins:** time **83.1** vs 74.1; rows **26.6** vs 16.5.

Why rows lie. The loop samples at **24–26 Hz** while actuating and **341–351 Hz** otherwise, because only actuating pays the serial round trip. A row count weights each state by how *fast* it samples.

The distortion is not constant: rows understate DAMPING by $3.1\times$ on v9 and $4.5\times$ on v10. Row-based figures are not comparable to *each other* either.

The fix, and the number that moved.
Integrate timestamps.

v10: merging the branches, and soft-saturation



The new fix. v9 faulted the whole rig after 30 consecutive pinned samples. But **clipping removes authority in one direction only**: an output pinned at v_{max} still pulls *down* at full strength. A clipped loop is a **weakened** loop, not a broken one, and it is usually the thing bringing the optic back.

$$\text{sat_trip} = (\text{sat_streak} > 30) \wedge \neg(\text{env} < 0.98 \text{env}(t - \text{LAG}))$$

The same level-against-trend correction runaway-trend made, one interlock over. Same history, same lag, so “falling” means one thing to both.

Why the merge needed a third change. A bias step is a force step: it rings the pendulum inside the band both trend tests read. So a step **invalidates** the history instead of suspending the interlocks, which would open a 16s hole every 15s.

v10 on the bench, 2026-08-06: bias-trim inside a live loop

trim step	total offset	verdict
ch3 → 0.50 V	576 → 560	kept
ch3 → 0.75 V	560 → 532	kept
ch3 → 1.00 V	532 → 507	kept
ch1 → 0.50 V	507 → 472	kept
ch3 → 1.25 V	472 → 564	reverted
ch1 → 0.75 V	489 → 477	kept
ch1 → 1.00 V	477 → 481	reverted
ch0 → 0.50 V	501 → 522	reverted

Total |offset from mid-scale|, four OSEMs, counts.

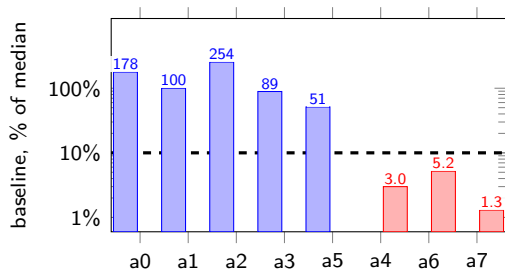
Accept-or-revert earns its keep. 3 of 8 steps made the *rig* worse and were frozen. ch3 at 1.25 V cost **92 counts** in one step: four coils, two DOF.

More faults, no thrash. 35.7 s of FAULT over 7 trips is **5.1 s each**, the mandatory FAULT_CLEAR_SUSTAIN_S, as v9's 5.05 s.

ch3 caused 7 of the 8 events (ch1 and ch2 one each). It has the **weakest coil** at −51 counts/V against ch0's −105 and ch2's +213, and it is the channel the trim stepped three times. **A bias step is a force step.** *Trim the weakest coil last.*

soft-saturation never fired: every trip was the runaway breaker. But auto-disable did, for the first time on hardware: ch2 railed, **3 of 4** kept damping, re-armed 2 s later.

v10: baseline-floor, the guard v5.5 needed



Dashed: $\text{BASELINE_FLOOR_FRAC} = 10\%$ of the median over *enabled* channels, 2026-08-06. **It sits in an empty**
 $10\times$ **gap**: 5.2% (worst dead) to 51.3% (a5).

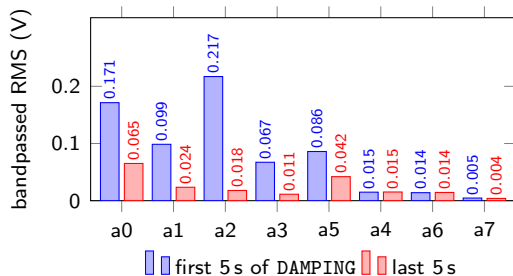
The rule. At the end of every calibration, a channel whose baseline is under 10% of the **median** across enabled channels is demoted exactly as auto-disable demotes a railed one.

Median, not mean, and that is the whole safety argument. The largest baseline is always $\geq$ the median $>$ the threshold, so **at least one channel always survives**: no quorum arithmetic. And if *most* of the rig is dead the median *is* a dead channel, the threshold collapses, and the guard does nothing, rather than demoting six on the word of two.

Relative, not absolute, because an absolute floor in volts would depend on the seismic background and each OSEM's alignment. On a quieter night a real channel and a dead one fall together.

And note what did *not* save v5.5. Those three were *config-disabled* in v10 and still had to be demoted: the runaway breaker gates on healthy, not on enabled. **Switching a channel off does not take it out of the interlocks.**

a5: a sign predicted out of noise, and then retracted



STEADY_GAIN[5] = -0.015 was a **prediction**: a5's DC diagonal is -7 counts/V against a block mean of 4, under $2\times$ the noise. **A wrong sign pumps.**

It damped: $0.0860 \rightarrow 0.0420$ V, while ch0 fell $0.1714 \rightarrow 0.0651$.

The three at zero gain are the control.

a4/a6/a7 are **flat to three decimals** across 40 s that took a factor of 3–12 out of every real channel.

Then the newer measurement retracted it. The fixed-frequency refit (2026-08-06) finds a5's residue **not distinguishable from its own noise**, and a gain on a channel that measures noise *injects* actuator noise while damping nothing.

v12 ships STEADY_GAIN[5] = 0. The two measurements disagree and **the one with error bars wins**. a5 is still in the quorum while nothing drives it, which is what holds the rig out of LOCKED.

v11: all eight, and the transport stops throttling the loop

1. sat-window: a bug fix, not a retune.

MAX_CONSECUTIVE SATURATED = 30 counted *samples*, so it meant 0.41 / 1.28 / 1.65 / **2.40**s at 1 / 4 / 5 / 8 coils, and one clean sample erased the whole streak.

Now SAT_SUSTAIN_S = 1.30 s with SAT_FRACTION = 0.80: the shape the *rail* check has always had. $30 \times 42.6 \text{ ms} = \mathbf{1.278 \text{ s}}$, so it is **behaviour-preserving on four channels**.

It goes **first**, because the other two changes both move the sample rate. *You do not move a sample rate underneath a sample-counted threshold.*

2. fast-transport. pyDAC2.FastDAC writes and returns with no ack wait, and the baud goes 115200 $\rightarrow$ 500000. **18 Hz $\rightarrow$ 1111 Hz.** Requires a **reflash**; a host at the wrong baud fails preflight with “never sent READY”.

3. All eight enabled, with a4/a6/a7 as *sensors* at **zero gain**.

baseline-floor demotes all three on the first calibration, and that is the designed outcome. ENABLE_CHANNEL stops being a constant somebody maintains and becomes a **measurement the rig makes every time it calibrates.**

And then it damped worse than everything before it.

240s, 3 kicks	DAMPING	FAULT	faults
v9 (4ch, 23 Hz)	83.1%	8.5%	4
v10 (8ch, 18 Hz)	74.1%	15.0%	8
v11 (8ch, 1111 Hz)	42.3%	25.3%	11

A $49\times$ faster wire made the damping worse.

Defect 7: a torn serial frame, and 45 V into a 0.5 V rail

The data. `read_sample()` accepts any row whose first character is a digit and which splits into $\geq N$ fields.

valid, every field in 0..1023

612,634,701,688,559,548,861,733

torn: 8 fields, first char a digit, **accepted**

612,634,**522676**,688,559,548,861,733

Where it comes from. fast-transport stopped draining the board's OK `ch=.. v=.. ack`, so acks now interleave with data rows and a buffer boundary splices two fields' digits together.

The speed-up created the defect. `v11`'s own 240 s log carries **ten** rows with a count outside 0..1023 (5659, 65690, 522676). **Zero** such rows appear in any of the **four** slower logs from the same session.

What goes wrong. Follow one sample through the loop:

	value	step
1	522676 counts	the torn field
2	$\times 5.02/1023$	= 2564.8 V raw
3	$\times \alpha_{LP} = 0.0178$	= + 45.6 V into bp
4	rings down, $\tau_{HP} = 0.398$ s	the 0.4 Hz highpass
5	$\hat{v}$ peaks	= 1401 V/s
6	$P = 0.030 \times 1401$	= 42 V demanded
	rail	0.5 V

$$\alpha_{LP} = \Delta t / (\Delta t + \frac{1}{2\pi \cdot 3 \text{ Hz}}) \text{ at } \Delta t = 0.96 \text{ ms.}$$

One bad character pins the actuator for the whole 0.4 s ringdown of the highpass. **Four of `v11`'s twelve entries to FAULT follow a torn line within 2 s**, and the first starts the 4 s-DAMPING / 5 s-FAULT thrash that eats the middle of the run.

v12: sample-guard and decimate, and why they ship together

The fix, and the number that moved.

sample-guard: a count outside

0..ADC_MAX_COUNTS is a **framing error**, dropped exactly the way an OK line is. Three lines.

Replayed offline over the same v11 log, ch0's velocity estimate:

ch0 velocity	v11	+guard
rms (V/s)	13.109	5.413
peak (V/s)	1441	41

A 35× reduction in peak, on data already recorded. No bench time, no hardware change.

decimate: give the loop its own clock. Read the wire at full rate; run the control step at CONTROL_HZ = 100 on the *mean* of whatever arrived since the last one.

Why: the arrival rate has been 24/12.5/18/1111 Hz, and every gain, filter and interlock here was tuned against one of them.

ch0 velocity rms (V/s)	with guard	without
v11	13.109	13.109
+ decimate	4.732	23.4

Without the guard, decimation makes it worse.

A boxcar mean *spreads* one bad sample across a whole step instead of confining it to one. **The two are not separable.**

Phase cost is measured: 100 Hz gives back 13.1° at 3.75 Hz against v11's wire, and still holds 18.4° *more* than the four-coil configuration every gain here was validated at.

Defect 8: the runaway breaker was reading the *beat*

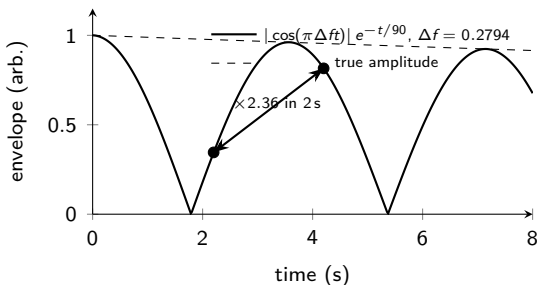
The data. Three modes, measured to **0.0005 Hz** on the 320 s quiet record. Beat period = $1/|f_i - f_j|$:

pair	$ \Delta f $ (Hz)	beat period
A-B	0.2794	3.578 s
B-C	0.6447	1.551 s
A-C	0.9241	1.082 s

$$f_A = 0.7155, f_B = 0.9949, f_C = 1.6396 \text{ Hz.}$$

RUNAWAY_TREND_LAG_S = 2.0 sits inside all **three**. It was chosen in v9, before any of these frequencies was known.

What goes wrong. $\text{env}(t)$ and $\text{env}(t - 2\text{s})$ land on *different phases of the beat*. The test measures the beat and calls it growth.



Toy model, the A-B pair. At $t = 2.2$ s the envelope reads **0.3455**; at $t = 4.2$ s it reads **0.8154**. The test asks for $> 1.02\times$ and gets **2.36** $\times$, while the *true* amplitude has **fallen 2.2%**.

Defect 8, other half: a real ringdown was invisible too

The data. τ_{env} , the decay of the *envelope the breaker watches* after a kick, measures **56–130 s**. How far does that fall in one lag?

	$\tau_{\text{env}} = 56 \text{ s}$	130 s
fall in 2 s	3.5%	1.5%
fall in 10 s	16.4%	7.4%
threshold: 2%		

What goes wrong. At $\text{LAG} = 2 \text{ s}$ a genuine ringdown moves **1.5–3.6%**, straddling the 2% line. **The breaker could not tell a real decay from a beat, in either direction.**

The fix, and the number that moved. `RUNAWAY_TREND_LAG_S` 2.0 $\rightarrow$ **10.0 s**: **2.8 $\times$** the longest beat, **5.6–13 $\times$** shorter than τ_{env} .

The remaining 14.4% is not thrash. $7 \times 5 \text{ s}$ of mandatory hold = 35 s = 14.3% of 244.5 s.

% of time	DAMP	FAULT	CALIB	entries
LAG = 2.0	54.1	25.2	20.7	10
LAG = 10.0	77.5	14.4	8.2	7
<i>whole suite, everything else identical</i>				
LAG = 2.0	43 passed / 7 failed			
LAG = 10.0	42 passed / 8 failed			

Two runs, 244.5 s, five hand kicks.

The fix is not free. One assertion flips: the **wrong-sign detector**. With ch2 inverted over 120 s, a 2 s lag trips the breaker ≥ 3 times and a 10 s lag trips it 2. The check demands 3.

The wrong sign is still caught: the companion faults > 0 passes at both. **A breaker that no longer reads a 1.55 s beat as growth also fires less often on something genuinely growing.**

Not a win over v9's 83.1%: different kick schedule.

Defect 9: stationarity is not quietness

The data. Caught on hardware: a fresh calibration produced a noise floor **11.5×** the trusted one.

What that does to every derived threshold, taking v3's ch0 floor as the trusted value:

	trusted	inflated 11.5×
baseline (V)	0.1804	2.075
runaway line, 1.8×	0.325	3.735
lock line, 0.35×	0.063	0.726

What goes wrong. The breaker goes blind and the rig declares LOCKED at **11.5×** the residual motion it should require. The calibration measured the *ringing*, not the room.

And persist-baseline would carry it across restarts.

The fix, and the number that moved.

baseline-sanity. A fresh floor that fails the check is **refused** and the trusted one kept. On the bench the 11.5×

 floor was refused.

And the guard needs a guard.

MAX_BASELINE_REFUSALS = 3 stops it refusing forever if the room genuinely *did* get louder.

What the persistence buys. The noise floor is a property of the room, not of the run, so v12 writes it to disk and reuses it across restarts, behind a config fingerprint, an age limit and a sanity check.

cold-start calibration	before	after
	20 s	2.0 s

This completes defect 3, nine versions later. v3 made one bad sub-window outvotable; v12 makes a bad *whole calibration* refusable.

Open defect 1: the rig locks, and cannot say so

The data. Zero *** LOCKED *** lines in every run of the 2026-08-06 session: v10, v11 and v12 have never printed it. The lock time is **the deliverable**.

But the rig does lock. The four `ch{i}_locked` columns are simultaneously 1 for **6 distinct episodes** on the good v12 run, and **3** on the run before it.

What goes wrong. The announcement needs `self.locked[live].all()`, where `live = enabled & healthy` and `ENABLE_CHANNEL` is `True` on *all eight*.

One undriven channel vetoes the whole rig, permanently. This is a quorum bug, not a damping failure.

	in live?	driven?	ratio	locked
a0	yes	yes	< 0.35	1
a1	yes	yes	< 0.35	1
a2	yes	yes	< 0.35	1
a3	yes	yes	< 0.35	1
a5	yes	no	1.26–2.53	0
a4	no (NOSIG)	no	n/a	n/a
a6	no (NOSIG)	no	n/a	n/a
a7	no (NOSIG)	no	n/a	n/a

a4, a6 and a7 demote themselves to NOSIG and leave `live`. **a5 does not**: it stays healthy, so it is in the quorum, but `STEADY_GAIN[5] = 0` so nothing drives it.

The fix, and the number that moved. Not yet applied. Drop zero-gain channels from the quorum. **One line**, and first on the open list.

Open defect 2: no OSEM rests at mid-scale

The ADC reads **0 to 1023 counts**, so **mid-scale is 511.5**. Room before a rail is **above** = 1023—resting, **below** = resting.

The data. Resting counts, CALIBRATING, 22 246 samples, outputs at bias:

	ch0	ch1	ch2	ch3
resting count	601.2	630.3	686.5	681.8
room above	422	393	336	341
room below	601	630	686	682

Every channel rests *above* mid-scale, so every channel meets its *top* rail first. ch2 is worst, with **2.0×** as much room below as above.

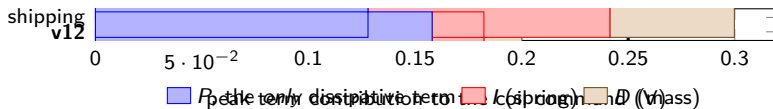
The fix, and the number that moved. **Proposed, not yet run: the bias sweep.** Coil bias is a static force, so step one channel 0.10 → 0.40 V at zero gain and log the resting counts. **If they move**, the offset is electromagnetic and a bias trim removes it; **if not**, it is mechanical flag alignment. **Two commands and a log.**

What goes wrong. A railed sensor is outside its linear region, so the velocity estimate from it is wrong **exactly when the optic is moving hardest and the loop is pushing hardest.**

of 251 280 samples	ch0	ch1	ch2	ch3
railed	2.9%	0.8%	3.6%	0.9%
rail interlock caught	136	0	51	0

The interlock needs the pin *sustained*, so it caught **187 of 20,696**. **About 99% of this clipping is invisible to every interlock.**

Open defect 3: the clipping is arithmetic, not bad luck



The data. Shipping gains, peak demand:

$$0.1279 + 0.1135 + 0.0584 = \mathbf{0.2998 \text{ V}}$$

against a half-window of **0.250 V**. **120% of budget.**

What goes wrong. The channels clip because the demand exceeds the window, not because of bad luck. And the largest item, K_i at **0.1135 V**, **38%**, dissipates *nothing*: the integrator sits behind the 0.4 Hz highpass, so $i = -K_i \text{ bp}$ and bp carries no DC to reject.

Nothing in this project justifies **KI_GAIN** or **KD_GAIN**. Not a scope trace, not a sweep, not a sentence. -0.030 has one. And v2 measured what they buy: **0.028** against v1's P-only **0.025**.

The fix, and the number that moved. v12 cuts K_i and K_d to 15% and raises K_p to the 0.035 ceiling.

	shipping	v12
peak demand	0.2998 V	0.2001 V
authority that <i>damps</i>	43%	88%

Still to be confirmed on the bench. v3 is the A/B baseline.

Four gains, two observable degrees of freedom

The data. Fit the plant from the dither run and ask what the added damping depends on:

free gains per term, ch0–ch3	4
observable degrees of freedom	2
undetermined dimension	2

Damping constrains exactly **two linear functionals** of a four-element gain vector.

So a two-dimensional family of gain vectors damps identically. Every direction inside it is indistinguishable by measurement, and the optimiser picks its point **by minimum actuator effort, not by measurement.**

This is ill-posedness, not a missing experiment. More dither does not shrink a null space.

Flat is as good as any point in that family. At the 0.035 ceiling, flat ± 0.035 reaches **90% of the achievable maximum**, worst-mode added decay 0.0739 s^{-1} .

The measurable lever is the actuator budget, not the gain shape.

Open defect 4: a file that refused to run, and did not

The data. `osem.v13.py` declares `KIND = "skeleton"` and its own docstring says it must not drive the optic.

```
before
def _unused_main(): raise SystemExit(...)
def main():         # runs v12's loop
```

```
after
def main(): raise SystemExit(...)
def _run_as_v12(): # deliberate path only
```

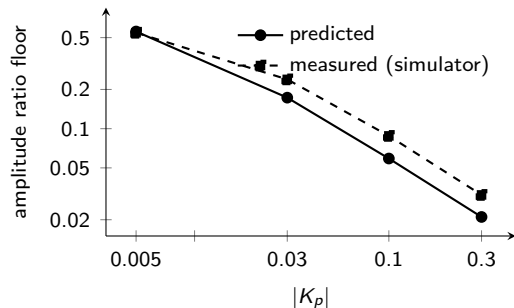
What goes wrong. The refusal was **defined and never called**. **make run V=v13 would have driven the optic** from a file whose own docstring forbids it.

The fix, and the number that moved. Swap the names. `main()` refuses; `_run_as_v12()` is the deliberate path.

A second one, equally invisible. `v13` does not define the module-level constants a controller must, so `S.load()` raised `AttributeError` on `ENVELOPE_WINDOW_S` and took make list down with it. Fixed by the `KIND` declaration, and `harness.declared()` now reads the `VERSION_TAG`, `BENCH_STATUS` tuple so a file that is never loaded can still report its own status.

And one still standing. `_run_as_v12()` does `v12.Controller = Controller`: it *mutates the v12 module*. That is not the shipping form. `bench.py` prints the gain vectors out of the file about to run, and a controller importing its body from another file defeats exactly that check.
Flatten it before treating it as a version.

The floor: why more gain is the only way down



The ratio settles at **0.23**, and that is the *steady state of a continuously forced oscillator*, not a failure to converge. Remove the drive mid-run and it decays with $\tau = 3.4$ s damped against 15.9 s gain-off (predicted 2.76 / 15.92).

The floor tracks $\frac{\text{intrinsic}}{\text{intrinsic} + \text{added}}$ damping, so it falls monotonically with gain.

The floor falls with gain, and gain is capped at $|K_p| = 0.035$ by the instability bound.

Those two facts are the vice the project sits in.

The residual **0.0089 V** is the actuator's 0.5 mV deadband and 10 ms throttle, *not* sensing: 0.0089/0.0087/0.0094 V at 20/5/0 mV of injected sensor noise. Fixing it moves 0.009 to 0.009 while you sit at 0.23.

What the simulator can and cannot tell you

It runs the *real* controller. Every version is imported and stepped, not reimplemented. Only `serial` and `DACController` are stubbed.

`SIM_CHANNELS = (4, 8)`, against a *measured* eight-OSEM body. **Four- and eight-channel controllers are both simulated.** Only `osem.v6.py` and `osem.v13.py` are skipped, both by `KIND`.

The plant is a linear second-order oscillator per mode with **exactly one nonlinearity**: the ADC, which offsets each channel to its resting count, rounds, and hard-clips at 0/1023. That is why clipping emerges on its own, with the bench's asymmetry.

What it cannot tell you:

- ▶ **A safe gain.** More negative K_p is monotonically more damping all the way to -0.6 . The ADC model reproduces *clipping*, not instability.
- ▶ **A wrong coil map.** It round-trips its held-voltage dict through whatever map the controller declares. **Only the bench catches one.**
- ▶ **A torn serial frame.** The transport is stubbed, so defect 7 could not have been found here.

make check: **220 passed, 11 failed** in 925 s.
sample-guard, decimate and all five
persist-baseline assertions pass. **The 11 failures are in warm-restart / baseline-reuse and are *not yet attributed***: only 2 of the 11 were ever read.

Still open, in the order I would do them

1. **LOCKED is never announced.** a5 vetoes the quorum at ratio 1.26–2.53 against 0.35 while nothing drives it. **One line**, and it is the deliverable.
2. **The bias sweep.** Step one channel $0.10 \rightarrow 0.40$ V at zero gain, log resting counts. Settles whether the 601/630/687/682 offset is electromagnetic or mechanical. **No code.**
3. **Confirm v12's gain cut on the bench.** Peak demand $0.2998 \rightarrow 0.2001$ V, dissipative share $43\% \rightarrow 88\%$.
4. **Cut LOCK_SUSTAIN_S = 5.0**, on its own: five of the ten-second budget confirming a lock that already happened.
5. **Gate fault re-engagement on amplitude**, not a fixed 5 s, which is 35 of v12's 35.2 fault-seconds.
6. **Kalman velocity estimator**, unblocked: f to 0.0001 Hz, $\tau > 138$ s so Q stops mattering, measurement noise measured.
7. **Read all 11 make check failures.** Only 2 were seen.
8. **a4, a6, a7.** Connected, near mid-scale, $\sim 1/100$ the in-band gain of a0–a3. **Cause not established.**

Learn the model, the estimator and the anomaly detector; keep the control law and the interlocks classical and certifiable.

The ladder, end to end

	channels	I/D	what it added	ch0 ratio	lock	bench
v0	ch0 only	zeroed	<i>the baseline</i>	0.188	12.1 s	2026-07-15 (scope)
v1	all four	zeroed	all four channels on	0.025	10.7 s	2026-08-03
v2	all four	live	real K_i , K_d	0.028	10.5 s	2026-08-03
v3	all four	live	all 3 defects fixed	0.029	10.6 s	validated
v4	all four	live	auto-disable	0.029	10.6 s	2026-08-04
v5	all four	live	fast-refault	0.029	10.6 s	2026-08-04
v5.5	<i>all eight</i>	live	eight-OSEM build	n/a	10 faults	08-04, no lock
v6	<i>bench tool:</i>		stepped-sine actuation matrix, closes no loop			08-04
v7	all four	live	bias-trim	0.029	10.6 s	validated
v8	all four	live	fast-calib, warm-restart	0.029	10.5 s	2026-08-04
v9	all four	live	runaway-trend	0.029	10.5 s	83.1% 08-04
v10	5 of 8	live	soft-saturation, baseline-floor	n/a	n/a	74.1% 08-06
v11	<i>all eight</i>	live	sat-window, fast-transport	n/a	n/a	42.3% 08-06
v12	<i>all eight</i>	cut	sample-guard, decimate, LAG	n/a	n/a	77.5% 08-06
v13	<i>skeleton:</i>		refuses to run			n/a

Ratios and lock times are **simulator** numbers; the percentages are **time-weighted** DAMPING on the bench. **v9's 83.1% and v12's 77.5% ran different kick schedules and are not comparable.** The control law did not change from v2 to v11: everything between is the supervisor around it. make check: 220 passed, 11 failed.